Algebraic Geometry Problem Set #1

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Problem (Exercise 1). *Let $f \in k[X]$ (with $\mathrm{char}(k) \neq 2$) be a polynomial of degree ≥ 1. Show that the plane curve*

$$Y = \{(x, y) : y^2 = f(x)\} \subseteq k^2$$

is irreducible, and that it is smooth if and only if $f(X)$ has no repeated roots. Show that

$$Y = \{(x, y) : y^2 = x^3 + ax + b\} \subseteq k^2$$

is smooth if and only if $4a^3 + 27b^2 \neq 0$.

Solution. *Part (i).* Recall that $Y = V(g)$ (with $g(x, y) := y^2 - f(x) \in k[x, y]$) is irreducible iff the coordinate ring $\mathcal{O}(Y) = k[x, y]/\sqrt{(g)}$ is a domain. Note that $y^2 - f(x)$ cannot be split into two factors in $k[x, y]$ unless f is not square-free, thus $\sqrt{(g)} = (g)$ is prime and Y is irreducible.

Let us now show that Y is smooth iff $f(X)$ has no repeated roots. Note that by definition for Y to be smooth one needs to have both $\partial g/\partial x$ and $\partial g/\partial y$ not vanish at any point of k^2. Note that the condition $\partial g/\partial y = 0$ implies $y = 0$, while the derivative w.r.t. x implies $f'(x_0) \neq 0$ for any value of x_0 such that $f(x_0) = 0$. However, this is exactly equivalent to f not having repeated roots.

Part (ii). From *part (i)* if suffices to study when $f_{a,b}(X) := X^3 + aX + b$ has repeated roots. Namely, if x_0 were such a repeated root, then

$$\begin{cases} x_0^3 + ax_0 + b = 0, \\ 3x_0^2 + a = 0. \end{cases}$$

From the second equation we have $x_0^2 = -a/3$. Substituting this expression into the first equation, we get $\frac{2}{3}ax_0 = -b$ or simply $x_0 = -3b/2a$. Thus, $9b^2/4a^2 = -a/3$, which exactly means $4a^3 + 27b^2 = 0$.

Problem (Exercise 2). *Show that $Y = \{(t, t^2, t^3) : t \in k\} \subseteq k^3$ is an irreducible algebraic set of dimension 1. Find generators of the ideal $\mathcal{I}(Y)$. Show that $k[X, Y, Z]/(\mathcal{I}(Y))$ is isomorphic to a polynomial ring in one variable over k.*

Solution. We want to write Y as a vanishing set of polynomial equations. For instance, $Y := V(y - x^2, z - x^3)$ should work. To show Y is irreducible, we want to show that the coordinate ring $\mathcal{O}(Y) = k[x, y, z]/\sqrt{(y - x^2, z - x^3)}$ is a domain, or equivalently that $\mathcal{I}(Y) = \sqrt{(y - x^2, z - x^3)}$ is prime. But clearly $(y - x^2, z - x^3)$ is a prime ideal: define map $k[x, y, z] \to k[t]$ by $x \mapsto t$, $y \mapsto t^2$, $z \mapsto t^3$; its kernel is precisely $(y - x^2, z - x^3)$ while the image is $k[t]$ which is a domain. Thus, $\mathcal{I}(Y) = (y - x^2, z - x^3)$, and moreover according to the map above, $k[x, y, z]/(\mathcal{I}(Y)) \cong k[t]$. It is then clear that $\dim(Y) = \dim(k[t]) = 1$.

Problem (Exercise 3). *Let X be a Noetherian topological space.*

(a) Show that every subset $Z \subseteq X$ is Noetherian in its induced topology.

(b) Show that X is quasi-compact.

(c) Show that if X is also Hausdorff, then it is finite and discrete.

Solution.

Part (a). Consider the descending chain $F_0 \supseteq F_1 \supseteq F_2 \supseteq \dots$ of closed subsets of Z. Each F_i is closed in Z, thus has a form $F_i = Z \cap C_i$ for closed $C_i \subseteq X$. Chain $C_0 \supseteq C_1 \supseteq C_2 \supseteq \dots$ stabilizes since X is Noetherian. But then by intersecting this chain with Z, we get that $F_0 \supseteq F_1 \supseteq F_2 \supseteq \dots$ also stabilizes.

Part (b). Suppose $\{U_\alpha\}_{\alpha \in I}$ is an open cover of X. Assume it is not finite. Then, for every finite subset $J \subseteq I$, one has that $Q_J := \bigcap_{\beta \in J}(X \setminus U_\beta)$ is a closed proper subset of X. Thus, we have found an infinite family of closed sets $\{Q_J : J \subseteq I, J \text{ finite}\}$ that forms a strictly descending chain, which contradicts the Noetherian property of X.

Part (c). Every subspace of X is Noetherian by (a) and consequently quasi-compact by (b). Since it is also Hausdorff, it is compact, and thus moreover closed. Thus, every subspace is closed and hence the topology is discrete. Since in such case $\bigcup_{x \in X}\{x\}$ is an open cover of X, we can find a finite subcover, meaning X is finite.

Problem (Exercise 4). *Let X be a topological space.*

(a) If $Y \subseteq X$ is a subspace, then $\dim(Y) \leq \dim(X)$.

(b) Give an example of a topological space X and a dense open subset $U \subseteq X$ with $\dim(U) < \dim(X)$.

(c) Suppose that X is irreducible and $\dim(X) < \infty$. *If $Y \subseteq X$ is a closed subset with* $\dim(Y) = \dim(X)$, *then $Y = X$.*

Solution.

Part (a). Let $Z_0 \subsetneq Z_1 \subsetneq Z_2 \subsetneq \dots \subsetneq Z_n$ be a chain of irreducible closed subsets of Y. Closure of an irreducible set is irreducible*, thus $\overline{Z}_0 \subsetneq \overline{Z}_1 \subsetneq \dots \subsetneq \overline{Z}_n$ is a chain of irreducible closed subsets of X. Notice that since $\overline{Z}_i \cap Z = Z_i$, $\overline{Z}_i = \overline{Z}_{i+1}$ implies $Z_i = Z_{i+1}$, and so the length of both chains are the same. Thus, $\dim(Y) \leq \dim(X)$.

Part (b). To be honest I had to look it up. The most straightforward example seems to be the Sierpinski space: the set $X = \{0,1\}$ with the topology $\{\emptyset, \{1\}, \{0,1\}\}$. Then, the subset $U = \{1\} \subseteq X$ works: it is a dense open subset of X of dimension 0 (while X has a dimension of 1).

Part (c). Assume $Z_1 \subsetneq Z_2 \subsetneq Z_3 \subsetneq \dots \subsetneq Z_d$ be a maximal chain of irreducible closed subsets of Y of length $d := \dim(Y)$. Suppose $Z_d \neq X$. Then $Z_1 \subsetneq Z_2 \subsetneq \dots \subsetneq Z_d \subsetneq X$ is a valid chain of irreducible closed sets of X of length $d+1$, which is not possible. Thus, $Z_d = X \subseteq Y$, implying $X = Y$.

Problem (Exercise 5). *Show that every two curves in the projective plane $\mathbb{P}^2$ intersect.*

Solution. Consider two curves $X = V_{\mathbb{P}}(f)$ and $Y = V_{\mathbb{P}}(g)$ (for homogeneous polynomials $f, g \in k[x,y,z]$). Assume it so happens that $X \cap Y = V_{\mathbb{P}}(f,g) = \emptyset$.

*Commutative Algebra course, Problem Set #2 :)

Then, according to projective Nullstellensatz[†], $\sqrt{(f,g)} \supseteq (x,y,z)$. Since (x,y,z) is a maximal ideal, $\sqrt{(f,g)} = (x,y,z)$. But notice that the dimension of $\sqrt{(f,g)}$ is at most 2 while (x,y,z) has a dimension 3. Contradiction.

[†]See https://ziyuzhang.github.io/ma40188/Lecture08.pdf, Theorem 4.24 for reference.